



Matteo Martini

Date of birth: 1994/11/21 | **Place of birth:** Sanremo, Italy | **Nationality:** Italian | **Email address:** matteo.martini.3ap@gmail.com | **Website:** www.mmartini.it

EDUCATION AND TRAINING

2023/11/01 – CURRENT Genoa, Italy

PHD IN COMPUTER SCIENCE University of Genoa

Development of innovative rehabilitation solutions and assessment tools based on Virtual Reality and unconventional interaction modalities (eye tracking, motion tracking), aimed primarily at individuals with severe motor and cognitive disabilities, as well as at milder clinical conditions such as anxiety disorders. The main application area is Rett syndrome, where motor and verbal impairments make conventional assessment and rehabilitation tools inadequate; moreover, interactive environments can foster levels of attention and engagement that are difficult to achieve with traditional approaches. Applications range from upper-limb motor rehabilitation to cognitive and educational interventions, and from visual assessment to communication support, either digitizing established rehabilitation methods or defining new ones. The integration of features based on state-of-the-art AI models also makes it possible to dynamically adapt content and difficulty to each user's abilities and progress, overcoming the rigidity of standardized protocols.

Supervisors: Manuela Chessa, Danilo Pani

Field of study Virtual Reality, Extended Reality, Human-Computer Interaction | **Level in EQF** EQF level 8

2020/09/01 – 2023/03/29 Genoa, Italy

MASTER'S DEGREE IN COMPUTER SCIENCE University of Genoa

Development of a system based on a ZED camera and YOLO for recognizing real-world objects and reproducing them within a virtual scene with consistent position and size. The software makes it possible to notify users of a virtual reality application of obstacles in the surrounding environment in a non-intrusive way; experiments also demonstrated its applicability to other areas, such as the manipulation of real objects paired with a virtual counterpart in order to leverage passive haptic feedback.

Advisors: Manuela Chessa, Fabio Solari

Field of study Virtual/Mixed Reality, Deep Learning, Stereopsis, Computer Vision | **Final grade** 110 cum laude | **Level in EQF** EQF level 7 |

Number of credits 120 | **Thesis** Dynamic obstacle avoidance in Mixed Reality - An approach based on deep learning and stereopsis

2017/09/01 – 2020/10/29 Genoa, Italy

BACHELOR'S DEGREE IN COMPUTER SCIENCE University of Genoa

Analysis of the π -calculus, a formal language for describing concurrent processes. The study examined its formalization from two complementary perspectives: the external view of the system, modeled through the reduction relation and structural congruence, and the internal view, modeled through the labeled transition system. The Harmony Lemma, which establishes their mutual consistency, was also examined.

Advisors: Francesco Dagnino, Elena Zucca

Field of study Information and Communication Technologies | **Final grade** 110 cum laude | **Level in EQF** EQF level 6 | **Number of credits** 180 |

Thesis Pi-Calculus: Un Formalismo Per Computazioni Distribuite

2008/09/01 – 2013/06/01 Imperia, Italy

HIGH SCHOOL DIPLOMA IN ACCOUNTING AND COMPUTER PROGRAMMING IIS G. Ruffini

Field of study Information and Communication Technologies | **Final grade** 83 | **Level in EQF** EQF level 5

WORK EXPERIENCE

VISITING PHD STUDENT – FRAUNHOFER HEINRICH-HERTZ-INSTITUT – 2026/05/01 – 2026/10/31 – BERLINO, GERMANY

Visiting research period at the Interactive & Cognitive Systems lab of the Fraunhofer Heinrich Hertz Institute (HHI), led by Sebastian Bosse, under the supervision of Paul Chojewski.

The work focused on extending TeMoRett, a system developed and validated by the group to promote upper-limb mobility in individuals with Rett syndrome and reduce the characteristic stereotypies without resorting to mechanical splints. The extension is an application inspired by the principles of Augmentative and Alternative Communication (AAC) that combines reaching and hand-eye coordination training with the practice of expressive skills: any flat surface becomes a communication board, configured by the parent or therapist, through which the child answers questions and makes choices by pressing pictogram cards. AI-based features also allow boards to be generated from a voice request by the therapist, reducing their workload and the wait time between one request and the next.

UNIVERSITY OF GENOA – GENOVA, ITALY

TEACHING ASSISTANT - AUGMENTED REALITY (AR) – 2026/03/01 – 2026/06/30

M.Sc. in Computer Science
Instructors: Manuela Chessa, Fabio Solari

Assisted the instructors in delivering the course, with a focus on lab sessions and student support.

TEACHING ASSISTANT – PROGRAMMING LANGUAGES – 2025/09/01 – 2026/06/23

B.Sc. in Computer Science
Instructor: Davide Ancona, Francesco Dagnino

Assisted the instructors in delivering the course, with a focus on lab sessions and exams. Supported and assessed students online through GitHub Classroom.

TEACHING ASSISTANT – FUNDAMENTALS OF COMPUTER SCIENCE – 2025/03/01 – 2025/06/30

B.Sc. in Biomedical Engineering
Instructors: Manuela Chessa, Francesca Odone

Assisted the instructors in delivering the course, with a focus on lab sessions and exams.

TEACHING ASSISTANT – PROGRAMMING LANGUAGES AND OBJECT-ORIENTED PROGRAMMING – 2024/09/01 – 2025/06/30

B.Sc. in Computer Science
Instructor: Davide Ancona

Assisted the instructors in delivering the course, with a focus on lab sessions and exams. Supported and assessed students online through GitHub Classroom.

TEACHING ASSISTANT – FUNDAMENTALS OF COMPUTER SCIENCE – 2024/03/01 – 2024/06/30

B.Sc. in Biomedical Engineering
Instructors: Manuela Chessa, Enrico Puppo

Assisted the instructors in delivering the course, with a focus on lab sessions and exams.

TEACHING ASSISTANT – PROGRAMMING LANGUAGES AND OBJECT-ORIENTED PROGRAMMING – 2023/09/01 – 2024/06/30

B.Sc. in Computer Science
Instructor: Davide Ancona

Assisted the instructors in delivering the course, with a focus on lab sessions and exams. Supported and assessed students online through GitHub Classroom.

RESEARCH FELLOW – 2023/08/01 – 2023/10/31

Research Fellowship: Interaction systems for VR-based cognitive exergames.

Supervisor: Manuela Chessa

INSTRUCTOR, UNIGE SENIOR – 2023/01/01 – 2023/05/10

Advanced Digital Skills Course (40 hours)

The course covered the informed use of smartphones and tablets, from hardware and software structure to connectivity and settings management, then extended to cloud services for data storage and sharing. Particular emphasis was placed on digital life: digital identity (SPID, the Italian public digital identity system, and CIE, the Italian electronic ID card), digital signatures, certified email (PEC), interaction with public administration services, and creating a personal blog. The course concluded with an introductory overview of emerging AI-based technologies.

ACADEMIC TUTOR – 2022/09/01 – 2023/04/30

B.Sc. in Computer Science

Afternoon tutoring sessions supporting first-year undergraduate students alongside their courses. Reviewed and clarified concepts covered in lectures, led guided problem-solving sessions, and ran review sessions in preparation for exams, particularly for Introduction to Programming, Algorithms and Data Structures, and Computer Architecture.

WORKSHOP INSTRUCTOR – ARCOS S.R.L. – 2025/03/03 – 2025/04/30

"Virtual Reality and Augmented Reality" Workshop
"A scuola di... futuro" Project – Molassana e Prato Comprehensive School, Genoa, Italy

Designed and taught a multi-session workshop for 8th-grade students. Introduced the fundamentals of programming and computational thinking through interactive lessons supported by educational tools such as EduBlocks, using hands-on examples with immediate feedback. In the second part of the course, students gained first-hand experience with Virtual and Augmented Reality headsets and immersive applications.

RESEARCH FELLOW – UNIVERSITÀ DI CAGLIARI – 2023/04/01 – 2023/06/30 – CAGLIARI, ITALY

Research Fellowship: Development of immersive and non-immersive virtual reality applications integrating sensors for motion analysis.

Supervisors: Giulia Sedda, Danilo Pani

COLLABORATOR – AQUA DB S.R.L. – 2014/11/30 – 2015/11/30 – IMPERIA, ITALY

Remote software installation and technical support; development of custom features requested by clients.

FOTOGRAPHER AND VIDEOMAKER – 2014/11/30 – Current – IMPERIA, GENOVA, ITALY

Photo and video production across a range of formats, including interviews, documentaries, music videos, and short films.

CONFERENCES AND SEMINARS

2026/11/07 – 2026/11/07 Battaglia Terme
Speaker at the AIRETT National Conference

Tailored Learning: Virtual Reality and Artificial Intelligence for Rett Syndrome

Link <https://www.airett.it/convegno-nazionale-airett5-6-e-7-novembre-2026/>

2026/10/08 – 2026/10/08 Bari (online)
Speaker at IEEE ISMAR 2026, XRehab workshop

Feasibility Evaluation of a Semi-Immersive Virtual Reality System with Wearable Physiological Monitoring for Gait Rehabilitation in Rett Syndrome

Link <https://sites.google.com/view/xrehab2026/home>

2026/03/25 – 2026/03/25 Daegu
Speaker at IEEE VR 2026

Immersive VR for the Assessment of Spatial Skills in Adolescents: Performance, Gender Effects, and Links to Computational Thinking

Link <https://ieeivr.org/2026/program/papers/#session-51>

2025/10/08 – 2025/10/08 Daejeon (online)
Speaker at IEEE ISMAR 2025, XRehab workshop

A Virtual Room for Rett Syndrome: Design and Implementation of an Exergame for Upper-Limb Rehabilitation

Best Long Paper Award

Link <https://sites.google.com/view/xrehab/home>

2025/10/04 – 2025/10/04 Cagliari
Poster presenter at SIAMOC National Conference 2025

Out-of-the-lab assessment of VR-guided reaching movements in Rett patients

Link <https://www.siamoc2025.com/programma/>

2024/06/05 – 2024/06/05 Torino

Speaker at IEEE GEM 2024

Immerse: Immersive environment for representing self-avatar easily

Link <https://edas.info/web/ieeegem2024/program.html#S1569628466>

2023/09/13 – 2023/09/13 Udine

Poster presenter at ICIAP 2023

Obstacle Avoidance and Interaction in Extended Reality: An Approach based on 3D Object Detection

Link <https://sites.google.com/view/iciap-2023/programme>

● SCUOLE E CORSI

2023/07/17 – 2023/07/21

European Agent Systems Summer School (EASSS)

School organized by the G²OAT research group under the auspices of EURAMAS (European Association for Multi-Agent Systems), with the support of ACM SIGAI and EurAI.

The program provided an overview of autonomous and multi-agent systems, structured around three main areas: design and programming paradigms for multi-agent systems; the formal foundations of agent reasoning and interaction—from computational logic to argumentation, with particular attention to their connection with explainable AI; and the algorithmic aspects of coordination and decision-making in competitive and cooperative settings. Lectures were complemented by hands-on sessions and a poster session.

2024/03/04 – 2024/03/08

Social XR Spring School (CWI, DIS)

Interdisciplinary program focused on Social XR as a medium for communication and collaboration. Sessions covered the capture and 3D modeling of realistic avatars and their behavior in virtual environments; the encoding, transmission, and distribution of volumetric video content, with particular attention to Quality of Experience (QoE) optimization techniques; ethics-by-design principles for developing responsible experiences; rendering and interaction techniques for immersive and multisensory applications; and human factors, along with methods for evaluating, modeling, and predicting QoE.

● PUBLICATIONS

2026

Adaptive exposure therapy for Social Anxiety Disorder: a preliminary validation in virtual reality

Social Anxiety Disorder (SAD) represents an increasingly prevalent challenge for healthcare. Exposure therapy is the elective treatment that has undergone significant development over last decades thanks to Virtual Reality (VR): its last frontier is represented by adaptive scenarios changing in real time to maximize efficacy and tolerability of the exposure. In the present study, 15 volunteers with clinical SAD symptomatology were immersed in two versions of the same virtual waiting room and asked to continuously report their anxiety in real time using the VR controller. One scenario gradually filled with avatars interacting with the participant in a fixed progression independent of the participant's responses; the other automatically adapted the avatars' number and behavior to induce more-or-less anxiety. Participants were not aware that, in the latter scenario, their ratings modulated the environment to maintain participants within an optimal range of social anxiety: sufficiently elevated to promote therapeutic habituation, yet low enough to minimize dropout. The current findings indicate that the adaptive condition successfully constrained anxiety within this target range: while preliminary, these results encourage further investigation to fully establish therapeutic efficacy.

Authors: Sergio Frumento, Noemi Paparo, Martina Briglia, Matteo Martini, Manuela Chessa, Alberto Greco, Danilo Menicucci | **Journal Name:** ACM ECCE 2026 | **Publisher:** ACM

2026

Multimodal Classification of Social Anxiety Using Continuous Self-Rating and Physiological Data in Virtual Reality

Assessment of Social Anxiety Disorder (SAD) is limited by paradigms that often target performance-related fears and its reliance on static self-reports that fail to capture the dynamic, in-the-moment nature of anxiety. This study introduces a novel Virtual Reality (VR) framework to assess SAD that integrates continuous self-ratings of anxiety with objective physiological data. Sixty-three participants were immersed in an ecologically valid VR waiting room scenario. They continuously rated their anxiety levels in real-time using a joystick, providing a direct subjective-behavioral measure, while their electrocardiogram and electrodermal activity were simultaneously recorded. We systematically compared three machine learning pipelines — a static Support Vector Machine, an end-to-end Transformer Encoder, and a Transformer Encoder on engineered features — to classify participants into the high and the low anxiety groups previously determined through the Liebowitz Social Anxiety Scale (L-SAS). To effectively harness psychophysiological and subjective data and overcome integration challenges, we developed a Late Fusion Transformer framework. This optimized architecture yielded the most robust and balanced classification (F1-score = 0.853 for the anxious class; 82.5% overall accuracy). The results show a synergistic benefit from combining modalities: while feature selection identified continuous self-rated anxiety as the

most powerful individual predictor, the late integration of physiological markers provided a significant complementary contribution that enhanced the model's overall accuracy, particularly in identifying non-anxious controls. In conclusion, our framework provides a dynamic assessment of SAD that captures the moment-to-moment experience of anxiety. By integrating a continuous self-rating of social anxiety with objective physiological data, this approach can identify clinically relevant incongruencies between a person's retrospective self-perception and their in-the-moment reactivity, paving the way for a new era of objective, personalized, and temporally sensitive psychometrics.

Authors: Marco Pardini, Sergio Frumento, Matteo Martini, Martina Alaimo, Gianluca Rho, Noemi Paparo | **Journal Name:** IEEE TAFCC | **Publisher:** IEEE

2026

Immersive VR for the Assessment of Spatial Skills in Adolescents: Performance, Gender Effects, and Links to Computational Thinking

Spatial skills, in particular mental rotation, are increasingly linked to STEM achievements and to computational thinking (CT) abilities. Traditional 2D spatial assessments face validity and equity issues from visual ambiguity, occlusion, and missing depth cues, requiring cognitively demanding 2D-to-3D interpretation. These artifacts might contribute to gender disparities, giving males an advantage that may reflect test bias rather than true ability differences. Immersive Virtual Reality (VR) seems to mitigate these issues by providing stereoscopic depth, multi-perspective viewing, realistic lighting, and embodied interaction. We compared spatial performance in VR and traditional 2D assessments using the Virtual Reality Mental Rotation Assessment (VRMRA) in a within-subjects study of 48 adolescents (ages 12–16). VR improved performance, yielding higher accuracy in Mental Rotation Test (MRT)-style tasks (+1.3 items, $p < .001$) and faster completion in Purdue Spatial Visualization Test: Rotations (PSVT:R)-style tasks (~100s, $p < .001$), while PSVT:R accuracy did not differ significantly between 2D and VR. We observed that gender effects are task-specific: VR improved the performance of females most in MRT-style tasks, reversing the 2D male advantage, whereas males gained more in PSVT:R-style tasks. Spatial scores correlated positively with CT across both 2D and VR assessments, indicating a link to computational problem-solving. These results suggest that the assessment of spatial skills in VR produces distinct performance patterns with respect to traditional 2D assessment, highlighting the importance of immersive VR for the development of improved tools and its potential implications for inclusive STEM education and talent identification.

Authors: Lorenzo Gerini, Matteo Martini, Giorgio Delzanno, Giovanna Guerrini, Fabio Solari, Manuela Chessa | **Journal Name:** IEEE TVCG | **Publisher:** IEEE

Integrating JaCa-Unity in VEsNA for Behavioural Realistic VR Simulations

Achieving high realism in Virtual Reality (VR) simulations with many Virtual Humans (VH) is a complex task and involves many different and challenging aspects. This paper focuses on enhancing VH autonomy to enable intelligent behaviours and interaction with the environment. Drawing from research in autonomous intelligent agents and Multiagent Systems (MAS), we analyse which results that the MAS community already obtained can be used in VR to improve VH behaviour, and which requirements should simulation tools fulfill in order to support them. To showcase our vision, we present the JaCa-Unity library for bridging the JaCaMo platform and Unity, and we show how JaCa-Unity has been integrated in VEsNA, an ecosystem that allows Jason agents to interact with humans via a chatbot-like interface and to be immersed within a VR environment.

Authors: Daniela Briola, Giuseppe Vizzari, Giuseppe Mirra, Andrea Gatti, Matteo Martini, Viviana Mascardi | **Journal Name:** Multi-Agent Systems | **Publisher:** Springer Nature

2025

A Virtual Room for Rett Syndrome: Design and Implementation of an Exergame for Upper-Limb Rehabilitation

Rett syndrome (RTT) is a rare neurodevelopmental disorder that severely impairs motor and cognitive functions, posing significant challenges for rehabilitation. This paper presents Virtual Room 2.0, a non-immersive Virtual Reality (VR) exergame designed to promote upper limb rehabilitation in RTT patients through engaging, interactive exercises. The system is based on marker-less body tracking obtained via a stereo camera, which allows its use in non-laboratory environments, such as homes and day centers. A 40-session, 10-week training protocol was conducted to assess the system's effectiveness, involving 10 RTT patients of different ages and clinical stages. Motion data were collected and analyzed to assess improvements in reaching movements. Results indicate a positive trend in upper limb function as well as increased motivation and engagement, confirming the potential of non-immersive VR-based rehabilitation for RTT patients.

Authors: Matteo Martini, Giulia Pili, Martina Semino, Michela Perina, Emanuela Riccio, Rosa Angela Fabio, Danilo Pani, Manuela Chessa | **Journal Name:** IEEE ISMAR 2025 - XRehab workshop | **Publisher:** IEEE

2025

Deep Learning-Based Classification of Social Anxiety Disorder Using Continuous Self-Reported Anxiety in Virtual Reality

Social Anxiety Disorder (SAD) is a mental disorder characterized by excessive fear and avoidance of social situations. Traditional assessment methods rely on retrospective self-reports, which may not fully capture moment-to-moment variations in perceived anxiety. To address this, we designed a novel Virtual Reality (VR) scenario to simulate a real-life social situation, specifically a waiting room that gradually fills with virtual characters. A continuous measure of self-reported anxiety was collected via joystick throughout the VR experience, allowing for real-time monitoring of subjective social anxiety. A one-dimensional convolutional neural network (1D-CNN) was trained to classify individuals with SAD based on their reported anxiety trajectories. The model was evaluated using a Leave-One-Subject-Out (LOSO) cross-validation strategy, achieving an F1-score of 0.82, recall of 0.89, and precision of 0.77, demonstrating strong classification performance. These findings suggest that self-reported anxiety alone is a viable signal for distinguishing individuals with SAD, paving the way for more accessible, sensor-free assessment tools in virtual environments. Future work will explore advanced feature extraction from the anxiety signal, integrate physiological markers, and investigate adaptive VR scenarios that dynamically respond to user-reported distress.

Authors: Marco Pardini, Sergio Frumento, Matteo Martini, Gianluca Rho, Valerio Vatteroni, Krishant Tharun, Martina Alaimo, Federico A Galatolo, Martina De Marinis, Enzo Pasquale Scilingo, Danilo Menicucci, Mario GCA Cimino, Manuela Chessa, Alberto Greco | **Journal Name:** MeMeA 2025
| **Publisher:** IEEE

2025

Virtual Reality as a Tool for Upper Limb Rehabilitation in Rett Syndrome: Reducing Stereotypies and Improving Motor Skills

Rett Syndrome (RTT) is a rare neurodevelopmental disorder that causes the loss of motor, communicative, and cognitive skills. While no cure exists, rehabilitation plays a crucial role in improving quality of life. Virtual Reality (VR) has shown promise in enhancing motor function and reducing stereotypic behaviors in RTT. This study aims to assess the impact of VR training on upper limb motor skills in RTT patients, focusing on reaching and hand-opening tasks, as well as examining its role in motivation and engagement during rehabilitation. Methods: Twenty RTT patients (aged 5–33) were randomly assigned to an experimental group (VR training) and a control group (standard rehabilitation). Pre- and post-tests evaluated motor skills and motivation in both VR and real-world contexts. The VR training involved 40 sessions over 8 weeks, focusing on fine motor tasks. Non-parametric statistical methods were used to analyze the data. Results: Results indicated significant improvements in the experimental group for motor parameters, including reduced stereotypy intensity and frequency, faster response times, and increased correct performance. These improvements were consistent across VR and ecological conditions. Moreover, attention time increased, while the number of aids required decreased, highlighting enhanced engagement and independence. However, motivation levels remained stable throughout the sessions. Conclusions: This study demonstrates the potential of VR as a tool for RTT rehabilitation, addressing both motor and engagement challenges. Future research should explore the customization of VR environments to maximize the generalization of skills and sustain motivation over extended training periods.

Authors: Rosa Angela Fabio, Martina Semino, Michela Perina, Matteo Martini, Emanuela Riccio, Giulia Pili, Danilo Pani, Manuela Chessa | **Journal Name:** Pediatric Reports | **Publisher:** MDPI

2025

Cooperation in Virtual Reality: Exploring Environmental Decision-Making through a Real-Effort Threshold Public Goods Game

We describe our contribution to the IEEE 3DUI Contest for IEEE VR 2025. Specifically, we present a collaborative Virtual Environment (VE) to be used as an innovative method to conduct experimental research in behavioral economics studies. Indeed, in such a field, analyzing collaborative behaviors is crucial, and collaborative VEs might be exploited to better control the variables of interest while preserving a high sense of realism and presence, to spread collaboration across wider geographical areas, and to maintain anonymity while preserving the sense of being with peers in-side environments. The developed system aims to be used as a general framework to perform a wide range of studies in the field, here explicitly focusing on a real-effort recycling task, where subjects decide whether to prioritize individual gains or contribute to a collective effort to meet a common recycling objective.

Authors: Manuela Chessa, Michela Chessa, Lorenzo Gerini, Matteo Martini, Kaloyana Naneva, Marianna Pizzo | **Journal Name:** IEEE VR 2025 - 3D UI Contest | **Publisher:** IEEE

2025

Immersive versus Non-Immersive Virtual Reality Environments: Comparing Different Visualization Modalities in a Cognitive-Motor Dual-Task

In fields like cognitive and physical rehabilitation, adopting immersive visualization devices can be unfeasible. In these cases, the main challenge is to develop Virtual Reality (VR) scenarios that still provide a strong sense of presence, usability, and user agency, even without full immersion. This paper explores a cognitive-motor dual-task in VR, consisting in counting and reaching, comparing three non-immersive visualization methods on a 2D screen (tracked perspective camera, fixed perspective camera, fixed orthographic camera) with the immersive experience provided by a head-mounted display. The comparison focused on factors like sense of presence, usability, cognitive load, and task accuracy. Results show, as expected, that immersive VR provides a higher sense of presence and better usability with respect to the non-immersive visualization methods. Unexpectedly, the implemented 2D visualization based on a tracked perspective camera seems not to be the best approximation of immersive VR. Finally, the two fixed camera conditions showed no significant differences in performance based on the type of projection.

Authors: Marianna Pizzo, Matteo Martini, Fabio Solari, Manuela Chessa | **Journal Name:** HUCAPP 2025 | **Publisher:** ScitePress

2024

Designing an Immersive Virtual Reality Scenario for Social Anxiety Elicitation and Modeling: A Preliminary Evaluation

We describe an immersive Virtual Environment (VE) designed and developed to record physiological and behavioral data to identify a computational model for real-time subjective anxiety estimation. The VE represents a waiting room with several chairs. Virtual Reality (VR) avatars come into the room and sit around the subject, sitting each in one of the available chairs, following two different strategies. Different forms of interaction between the avatars and the subject have been implemented. This paper presents preliminary tests of the VE conducted with 24 healthy volunteers self-reporting possible symptoms of Social Anxiety Disorder, who were invited to continuously report their subjective level of discomfort through the VR controller. Once integrated with the answers given to questionnaires aimed at investigating anxiety states and immersiveness in the VE, these preliminary results demonstrated the effectiveness of the proposed scenario as a novel and useful tool for robustly eliciting and assessing states of social anxiety in a realistic but nevertheless controllable and acceptable way.

Authors: Matteo Martini, Eros Viola, Francesco Bossi, Sergio Frumento, Alessio Iannizzotto, Sara Said, Alejandro Luis Callara, Fabio Solari, Enzo Pasquale Scilingo, Alberto Greco, Danilo Menicucci, Manuela Chessa | **Journal Name:** MetroXRaine | **Publisher:** IEEE

2024

IMMERSE: IMMersive Environment for Representing Self-Avatar Easily

Recently, the importance of having a representation of the users inside Virtual Reality (VR) by using avatars, which replicate the movements of real humans, has grown in several fields of application. Nowadays, developers can use many off-the-shelf devices for visualizations and tracking. Game engines like Unity3D and Unreal allow to easily integrate packages to manage head-mounted displays and other tracking devices. However, it is still difficult to combine different solutions and achieve a full representation of the human body, e.g., combining full-body and hand tracking or switching among different tracking modalities in an easy way. This paper describes IMMERSE, an open-source framework based on Unity3D, the XR Interaction toolkit, and an inverse kinematics solver. IMMERSE allows developers to insert a VR avatar in a virtual environment, animate it through different sources of 6DOF tracking measurements, which are also interchangeable during the simulation, and finally record the movements to animate other avatars or input to motion analysis techniques.

Authors: Eros Viola, Matteo Martini, Fabio Solari, Manuela Chessa | **Journal Name:** IEEE GEM 2024 | **Publisher:** IEEE

2024

Extended Reality and Artificial Intelligence for Exergaming: Opportunities and Open Challenges for Rehabilitation and Cognitive Training

The increased interest in the use of Virtual, Augmented, and Mixed Reality technologies, also referred to by the wider term eXtended Reality (XR), to create effective and innovative serious games and exergames in the healthcare domain opens several questions and points out open and still unsolved issues. In this paper, we will describe the main available software and hardware technologies and discuss the opportunities and open challenges in rehabilitation. We will also discuss the need to fill the existing gap between the advancement of techniques in the XR domain and their effective use in healthcare. Specifically, we will describe some use cases recently developed and included in clinical trials.

Authors: Manuela Chessa, Lorenzo Gerini, Razeen Hussain, Matteo Martini, Marianna Pizzo, Fabio Solari | **Journal Name:** IEEE RTSI | **Publisher:** IEEE

2023

Obstacle Avoidance and Interaction in Extended Reality: An Approach based on 3D Object Detection

In recent years, the compactness and capabilities of modern head-mounted displays have rekindled interest in Virtual Reality. There are now various options available, from discrete to integrated all-in-one solutions that no longer require an expensive workstation for graphical rendering. This has made it possible for a larger audience to own a personal VR device.

Despite manufacturers' safety recommendations, the home use of these devices can be dangerous. The currently available devices are equipped with basic safety systems that require defining a safe zone and just warn users whenever they are going to move outside of it. However, these systems are not capable of detecting obstacles in a realistic home environment. For example, living rooms or bedrooms may be cluttered with furniture and other objects that are not easily removable or that users will not remove, making it easy for them to hit something or trip over them.

To address this problem, we propose an obstacle avoidance system that combines state-of-the-art object detection with depth perception capabilities granted by stereopsis in RGB-D cameras. Our system can detect specific classes of objects, estimate their position and extent, and create a virtual counterpart for each one of them in an immersive virtual scene. This will help users visualize obstacles in their environment and avoid collisions.

Authors: Matteo Martini, Fabio Solari, Manuela Chessa | **Journal Name:** ICIAP 2024 | **Publisher:** Springer Nature

OTHER

2025/09/18 – 2025/09/21

Field Researcher at AIRETT Full Immersion 2025

Four-day, non-residential edition focused on field testing of ongoing projects. Sessions with end users covered: a cognitive application for reading and writing skills; visual ability assessment; assessment of preferences and level of realism; a sensorized treadmill coordinated with a semi-immersive VR environment; and physiological data collection to build a dataset for controlling an active orthosis designed to interrupt stereotypies.

2024/08/26 – 2024/08/30

Field Researcher at AIRETT Summer Camp 2024

Second participation. Field testing of the non-immersive VR environment and of a new eye-tracking-based cognitive application for reading and writing skills (object-word matching; word, syllable, and letter recognition).

2023/08/29 – 2026/08/31

Field Researcher at AIRETT Summer Camp 2023

Residential week organized by AIRETT, bringing together families of girls and young women with Rett syndrome, a multidisciplinary clinical team, and the university research groups collaborating with the association. The goal of my participation was field testing, with end users, of a non-immersive Virtual Reality environment developed to stimulate upper-limb movement and reduce stereotypies, collecting data and direct feedback on usability and effectiveness.

2026/10/06 – 2026/10/06

Organizer, XRehab Workshop (IEEE ISMAR 2026)

Workshop on the application of Extended Reality (XR) to rehabilitation: physical therapy, cognitive rehabilitation, motor recovery, pain management, and mental health. The workshop reviews the state of the art and future research directions, bringing together XR developers, healthcare professionals, and researchers.

Link <https://sites.google.com/view/xrehab2026/home>

2026/05/28 – 2026/05/29

Organizer, Computer Science Workshop 2026

The Computer Science Workshop (CSW) is an annual conference organized by PhD students and hosted by DIBRIS (Department of Informatics, Bioengineering, Robotics and Systems Engineering) at the University of Genoa, free of charge and open to all. It features talks by international speakers active in various fields of computer science research, presentations by local companies, and a poster session by PhD students. It is an engaging scientific and social event for students, faculty, and guests.

Link <https://docs-dibris.github.io/docs/Computer Science Workshop/csw26.html>

2025/05/29 – 2025/05/30

Organizer, Computer Science Workshop 2025

Link <https://docs-dibris.github.io/docs/Computer Science Workshop/csw25.html>

2024/09/09 – 2024/09/10

Organizer, Computer Science Workshop 2024

Link <https://docs-dibris.github.io/docs/Computer Science Workshop/csw24.html>

LANGUAGE SKILLS

Mother tongue(s): **ITALIAN**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
INGLESE	C1	C1	C1	C1	B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user